Fiscal Year:	FY 2007	**Task Last Updated:**	FY 03/26/2008
PI Name:	Tsotsis, Theodore Ph.D.		
Project Title:	The Study of Reactive Separations under Microgravity Conditions		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--In situ resource utilization		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
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No. of PhD Candidates:	5	**No. of Master' Degrees:**	
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Flight Assignment:	NOTE: End date changed to 9/30/2007 per PI (11/06) NOTE: Added to Task Book in 10/2006 per GRC		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Sahimi, Muhammad (USC)		
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Task Description:	The key project objective is to study the application of membrane-based reactive separation (MRS) technology under reduced gravity conditions. Such technology will be potentially of value for the production of propellants and fuels under reduced and microgravity conditions, but also in general for the development of effective chemical processing units under such conditions. The emphasis in this project is on the Sabatier reaction; this is because extensive efforts have been devoted to this reaction, in the context of regenerative/air revitalization processes, within the Space Exploration research program using conventional technology; this conventional technology may then serve as the "baseline" for proving the MRS technology. Successful development of the technology, however, will open the horizon for the MRS technology to a multitude of other applications relevant to in-situ development of fuels and other materials utilizing the largely CO2 Martian atmosphere and water resources found on Mars or brought from earth. Though some experimental work with macroporous membranes has been carried-out under reduced gravity conditions, we know of no application of membrane-based reactive separation technology under microgravity conditions. Under these conditions there is clearly a lack of integrated experimental/theoretical/computational studies of sorption and transport in membranes and membrane reactors under both non-reactive and reactive conditions. We, therefore, undertook in this project a systematic study of these phenomena. The necessary research was broadly divided into two parts. First the study of the membrane itself, and second the performance of the membranes in MRS applications for the Sabatier reaction. The various Tasks are briefly described below: Task 1. Development of High Temperature Membranes These membranes were prepared by depositing properly selected Layer Double Hydroxide (LDH) materials with CO2 affinity onto porous membranes substrates. In this Task these membranes were tested in order to establish that they are hydrothermally and chemically stable, and resistant to sintering under the proposed gaseous environments. These characteristics are important in determining whether these membranes and the MRS units based on them are likely to become an important part of the next generation of Air Revitalization Systems (ARS). The experiments are coupled with molecular simulations of the membrane structure, and its interactions with molecules of interest. Such simulations incorporate a broad spectrum of molecular mechanics and dynamics methodologies. The emphasis has been on utilizing the appropriate force-fields in order to simulate the sorption and transport characteristics of these membranes, and to demonstrate their applicability to the MRS technology. In addition to the LDH membranes, and in parallel, we also investigated the applicability of a different type of membranes, made of SiC. These membranes are known to have good hydrothermal stability. The goal in this Task was improving their separation characteristics, so that they can also be utilized in the Sabatier environment, and also validating their stability in this environment. Once the permeation performance of these membranes was investigated and validated under non-reactive conditions, we proceeded to Task 2, where a select series of Sabatier-MRS (SAB-MRS) experiments was carried out, as described below, to establish the ability of such membranes to perform satisfactorily under Sabatier-relevant conditions. Task 2. the Development of the Sabatier Membrane Reactor The objective of this Task is the design, development, and testing of the high temperature SAB-MRS system. To accomplish this goal, we undertook an extensive series of bench-scale testing of the concept. We investigated the effect on reactor performance of the mode of reactor operation and the choice of operating conditions. Experimental efforts in this Task were coupled with parallel modeling investigations intended to understand the important factors, at each length-scale, that control the ability of these reactor systems to affect transport, reaction and separation in fluid mixtures. These models were then be utilized in Task 3 focusing on Overall Process Design. Task 3. Overall Process Design In this Task we studied the technical feasibility and viability of the SAB-MRS. The primary emphasis was on the process design of the SAB step, as it is critical that one properly matches membrane and catalyst characteristics, in order to properly exploit the key features of both, the key goal being choosing the optimal set of operating conditions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The study of the application of the MRS technology to the Sabatier reaction represents a new research problem with great scientific and research interest. CO2 capture to recover its oxygen content, in the context of regenerative air revitalization processes for applications relevant to long duration missions in space, represents an ideal environment for the demonstration of this CO2 selective membrane-based reactive separation technology. Successful demonstration of the high temperature membranes developed for this application will open, however, the horizon to a multitude of other applications relevant to processing and use of Martian resources for in-situ development of fuels and other materials. The deliverables of this project is a comprehensive report of the factors that determine optimal performance for the SAB-MRS, particularly for space applications. Membrane and overall MRS unit performance in such an environment is key to the overall success, and this study undertook a systematic study of these phenomena.
Task Progress:	A key goal of our studies was the preparation of affinity-type CO2-selective membranes. In the preparation of these membranes, we have used a number of hydrotalcite sources, prepared in our laboratories and also available commercially; we have also used a variety of mesoporous and macroporous supports, including HT porous disks prepared in our laboratories, and alumina tubes and disks. We have also prepared two types of membranes, large area membrane disks and tubes, and micromembranes prepared on stainless steel foils and silicon wafers. The micromembranes show good potential for application in micro-fuel cells. The membranes have been tested for their transport characteristics using both single gases and mixtures of gases, as well as by a variety of other characterization techniques including SEM and TEM, DRIFTS, EDX, and DTA/TGA. Quality nanoporous membranes have been prepared, which show significantly higher permeation rates for gases with smaller kinetic diameters like He (used here as a safe surrogate gas for hydrogen) as compared to gases with larger kinetic diameters like Ar. Some of these membranes are permselective towards CO2. The effect of preparation conditions on the membrane transport characteristics have also been studied. Coating these membranes with a silicone layer improves their separation characteristics making them significantly more permeable towards CO2, as one would have expected based on the strong affinity of these materials towards CO2. In this study, we have also investigated a continuous regenerative approach which, in addition to removing the CO2, may potentially also allow for the recovery of oxygen. It involves the use of the methanation (Sabatier) reaction, in

	which the CO2 reacts catalytically with hydrogen to simultaneously produce methane and water. We have investigated, in particular, the application of a reactive separation technology, in which the catalytic and separation steps are coupled in situ through the use of high-temperature membranes. Coupling reaction and separation provides added synergy, which enhances the performance of both steps. Task 2 consisted of experiments, which involved catalyst and membrane characterization, and which were used to validate experimentally a reactor model. The model was subsequently utilized in Task 3 in order to investigate and establish the feasibility of the proposed reactive separation application for life-support systems.
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